Psy 111-Lecture notes**

INTRODUCTION TO PSYCHOLOGY

Main objectives:

1. What is psychology?
2. History of psychology
3. Psychology as a science
4. What are the major psychological perspectives?
5. Methods in Psychological investigationa

So What is Psychology?

Psychology is the science of human behavior and mental processes. It is the scientific study of behavior and mental processes and how they are affected by an organism's physical, state, mental state, and external environment.

Most people think of psychology as the study of differences between people, but it also includes the study of similarities between people.

The term psychology comes from the Greek roots *psyche* meaning soul or mind and *logos* meaning word or study

Why study Psychology?

1. Psychology helps us scientifically evaluate common beliefs and misconceptions about behavior and mental processes.
2. To have a knowledge base – to know what has already been investigated and understood.

3. To understand how theoretical developments have evolved in psychology through competition and opposition of ideas

Basic goals of scientific psychology

a. describe, b. explain, c. predict, and d. change/influence or control behavior and mental processes

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It must be noted that psychological information is based on *empirical evidence*. This is information based on direct observation and measurements with scientific method

How does the scientific method work?

The classical approach to solving anything in science starts with a "problem," which through experimentation and prediction, evolves through the "hypothesis" and "theory" stages into a scientific "law."

The **scientific method** consists of the orderly, systematic procedures that researchers follow as they identify a research problem, design a study to investigate the problem, collect and analyze data, draw conclusions, and communicate their findings

What is Scientific Thinking in Psychology?

It involves the following:

1. Critical thinking - assess claims on the basis of **well-supported reasons and evidence** - not on emotional or anecdotal reasoning.

2. It involves asking questions - one of the most important is, “WHY?” HOW? WHAT?

3. It involves defining terms – which must be clear and concrete

4. It involves examining evidence - “ I may have my opinion!” but it doesn’t count

5. It Involves analyzing assumptions and biases - scientific thinkers do not take anything as proven fact and work hard to overcome their own biases in thinking

6. It involves avoiding emotional reasoning - do not let gut feelings replace clear thinking - emotional conviction does not settle arguments

7. It involves avoiding oversimplification - the obvious answer is often wrong and misleading - do not argue based on own anecdotal evidence

8. It involves consideration of other interpretations - the best interpretations are supported by the most evidence and explain the most variables

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9. It involves tolerating uncertainty - sometimes evidence is unclear or does not even exist

10. involves asking questions that can be tested in this world

Brief History of Psychology

Why study history of psychology?

- To make connections in an incredibly diverse field of study.
- To discover interesting personalities or 'characters' in its history.
- To be able to critique our current thinking about psychology today. For example: Is the current "mindset" the most appropriate that will be refined and augmented in the years to come or will it be replaced?
- To understand the ethical issues pertaining to psychological questions.
    - Emotional distress for the participants of certain studies
    - Theoretical understanding vs. real-world application
    - To change behaviour or leave unadulterated

The History

- Aristotle- born in 384 BC was the first known writer in the field of psychology. Aristotle used the term "psyche" to refer to the essence of life. This term is translated to mean "soul" or "mind", but it is closely linked in meaning to the word "breath"

How did psychology evolve as a discipline

The founders of psychology were philosophers – the ancient Greeks were important antecedents to modern Western ideals largely influenced by Aristotle versus Plato- Therefore psychology has root in philosophy

- Plato- between 5th – 4th century BC Plato had raised issues of psychological improtantce. For example, what is knowledge, what is love? Is there a relationship between our thoughts and our behaviour? Plato referred to as a dualist –Mind/body

- Aristotle- who was Plato student- may be referred to as originator of psychology because he was the first to discuss a fundamentally a psychological issue. He discussed topics like sensation, memory sleep etc.. he also talked about systematic observation.

Major influences

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- Roman pragmatism and scepticism towards science and philosophy.

The renaissance ('rebirth') of science 14th – 16th century in Europe. This included Rene Descartes (1596-1650) he postulated on the 'doctrine of innate ideas' we are endowed from birth with inborn ideas. Dualist like Plato

Dualism - the idea that the mind and body are separate.

Body – physical, material, follows the physical laws of nature. Can be studied scientifically.
Mind – non-physical, spiritual, does not follow physical laws and therefore can not be studied scientifically

- 17th century – Psychological idea emerges as an outgrowth of empirical and rational philosophy
- Science displaced religion as an authority on understanding
- 18th century – Psychology began to emerge as a science in conjunction with studies of the nervous system
- Founder of Experimental Psychology: Wilhelm Wundt (1832-1920).
    - Established the first laboratory for the study of psychology in Leipzig in 1879;
    - Campaigned to make psychology an independent discipline;
- Wundt used scientific methods to study fundamental psychological processes, such as mental reaction times in response to visual or auditory stimuli. Example: He measured how long it took a person to detect the sight and sound of a bell being struck.
- In 1874, he published *Principles of Physiological Psychology*. Outlined the connection between physiology and psychology. It promoted his theory that psychology is a separate scientific discipline that requires experimental methods to study mental processes.
- The *Principles* advocated a systematic investigation of the immediate experiences of consciousness, including feelings, emotions and ideas, mainly explored through Wundt's system of "internal perception", or the self-examination of conscious experience by objective observation of one's consciousness.

When he established a lab at the University of Leipzig in Germany in 1879. Many psychologists and historians consider this to be the formal beginning

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of psychology as a scientific study. Wundt was responsible for creating the first school of psychological thought called *Structuralism*

- **Structuralism**, the first major school of thought in psychology, maintains that complex conscious experiences can be broken down into elemental structures or parts of sensations and feelings.- Wundt and Wundt studied consciousness using a method of looking inward at one's own experiences = called introspection.
- This group of Psychologists used a technique called *Introspection* which was a process of self-examination where the person described and analyzed thoughts as they occurred.
- Psychology was not considered an area of study until the 1870's when the ideas of the industrial revolution were applied to the fields of philosophy and physiology.
- The German scientist Wilhelm Wundt (1832-1920) opened the first laboratory and faculty of Psychology in the University of Leipzig in 1879 and established the first Psychological Journal in 1881. Founder of experimental psychology.
- He is estimated to have written over 54,000 pages of work He was the first to apply the scientific method to the field.
- Campaigned to make psychology an independent discipline.
- Defined psychology as the scientific study of conscious experience.

Structuralism vs. Functionalism

- At this time, psychology split into two main schools: structuralism vs. functionalism.
- Structuralists were led by Edward Bradfor Titchener (1867-1927), they believed that the task of psychology was to analyze consciousness into its basic elements and study how these were related. These elements would include ideas like sensations, emotions, and images.
- These investigations were based on introspection by trained subjects.
- Introspection – careful, systematic self-observation of one's own conscious experience.

Granville Stanley Hall (1844-1924)

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One of Wundt's students was G. Stanley Hall who was vital for bringing the science of psychology to America. He founded the American Psychological Association and became its first president, he started America's first research laboratory at John's Hopkins University in 1883, he began the first journal of psychology in America.

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William James

- Functionalism became popular with the publication of *Principles of Psychology* in 1890 by William James. (1842-1910)
- The work became the standard text for psychological departments and is still required reading in many university programs.
- Functionalism is the idea that psychology should focus on the function or purpose of consciousness rather than its structure.
- **Functionalism** is an early school of psychology that emphasized studying the purpose, or function of behavior and mental experiences. It looks at how behavior functions to allow people and animals to adapt to their environments. William James taught the first course on psychology and in 1890, published the first psychology text. William James was influenced by Charles Darwin.
  Functionalism is the second perspective to emerge, founded by William James.They studied how and why the mind functions. This perspective broadened the scope of psychology and applied psychology to practical settings. This perspective still focused on consciousness but began to examine overt behavior, not just mental processes.

Natural Selection of the mind

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- Functionalism is heavily based on the work of Charles Darwin and his theory of natural selection.
- Natural selection posits that heritable characteristics that provide an organism with a competitive advantage are more likely to be passed on to the next generation and thus less advantageous characteristics become extinct over generations.
- The Functionalists like William James applied this theory to humans. By studying what consciousness does for humans, we can discover how it works.

Comparison

- Structuralists focused their work in the laboratory and on sensation and perception.
- Functionalism took the field in new directions and focused on mental testing, education, and adolescent development.
- Functionalism's practical application of early psychological principles led to the formation of the next major development.

Psychological Perspectives

- Psychoanalytic perspective by Sigmund Freud.
- Where previous theories on psychology focused on conscious experiences, Freud looked inward. He sought to uncover causes of behavior that were unconscious, or hidden from the person's conscious awareness.
- Psychoanalysis emphasized the role of unconscious conflicts in determining behavior and personality.
- He believed that glimpses of the unconscious could be revealed in dreams, memory blocks, slips of the tongue and humor.
- Also, psychological disorders could result when unconscious conflicts became extreme.
- Free association and dream interpretation were developed technique to explore the unconscious processes

Behaviorist perspective

- Behaviourism was founded by John B. Watson in 1913.
- It stated that scientists should only study observable behaviour and that consciousness should be abandoned because ultimately, consciousness and perceptions are private events and cannot be objectively verified.
- Behaviourism sparked the nature vs. nurture debate with its strong emphasis on stimulus and response (S-R psychology)

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- Behaviour – any overt and observable response or activity by an organism.
- Stimulus – any detectable input from the environment.
- Major historical characters associated with this perspective are Ivan Pavlov and B F Skinner

B.F. Skinner

- Behaviourism gained greater influence in the 1950's with the work of B.F. Skinner, who was voted the most important influence in the history of psychology
- By using stimulus and response pairings, he could control the behaviour of rats and pigeons with remarkable consistency.
- He discovered the principle that organisms tend to repeat responses that lead to positive outcomes and not repeat responses that lead to negative ones.
- This simple principle would go on to form the foundation of behaviour modification used in schools, prisons, factories and hospitals.
- Skinner's work sparked heated debates over the concepts of free will and determinism in his work "Beyond Freedom and Dignity". If all our behaviour was determined by external stimuli, what choice did we really have concerning our behaviours and desires?

Humanistic perspective

- Humanistic Psychology emphasizes each person's unique potential for psychological **growth and self-direction**.
- Self-determination, **free will** and the importance of choice are important in psychological growth.
- The major advocates of this perspective are Carl Rogers & Abraham Maslow

Humanists revolt

- This debate sparked the next wave of innovation in psychology.
- Humanism was a direct reaction to the stripping of free will that psychoanalytic theory and behaviourism entailed.
- Humanism is the approach that emphasizes the unique qualities of humans, personal freedom, and personal growth.

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- A *humanistic psychology* argues that we can predict behaviour, not through deterministic laws, but based on common culture, values, language, development and attitudes.

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Carl Rogers and Abraham Maslow

- Humanism was championed by Carl Rogers and Abraham Maslow. They both believed that human behaviour is determined by an individual's "self-concept", which animals presumably lack.
- Each of these self-concepts seeks to grow, evolve and develop and psychological disturbances arise when these drives are blocked.

Cognitive perspective.

Ψ Cognitive psychology believes that behaviors are performed because of ideas and thoughts.
The cognitive perspective focuses on such processes as perception, memory, and thinking
That's what this area of psychology believes and Cognitive psychology currently exerts a strong influence in psychology.

Biopsychological perspective

Ψ How biological factors affect mental processes and how the brain effects behavior.
Ψ Behavior and biology interact in important ways, and we will discuss the impact of this field when examining psychopharmacology, development and genetics.

Social psychological perspective

Ψ Focuses on interpersonal behavior and the role of social forces in governing behavior. Typical topics include attitude formation, attitude change, prejudice, conformity, attraction, aggression, intimate relationships, and behavior in groups.

Evolutionary perspective

Ψ Applying the principles of evolution to explain psychological processes and phenomena
Charles Darwin

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- Wrote *On the Origin of Species by Means of Natural Selection*, his first book on evolution, in 1859.
- **The Theory of Evolution** -proposes the idea that individuals fight for survival
    - Species change over time and space.
    - All organisms share common ancestors with other organisms.
    - Evolutionary change is gradual and slow

Charles Darwin (cont.)

- In *The Voyage of the Beagle* Darwin formed his theory of natural selection by observing animals while traveling the world.
- **Natural selection** -The process whereby organisms better adapted to their environment tend to survive and produce more offspring (four main components)
    1. Variation
    2. Inheritance
    3. High rate of population growth
    4. Differential survival and reproduction

Methods in Psychology

The Scientific Method

systematic procedure of observing and measuring phenomena to answer questions about **What** happens, **When** it happens, **What** causes it, and **Why?**

- Formulate a research question
- State the hypothesis (People who report poorer self-images also score higher on a depression scale).
- Test the hypothesis (respond to statements: "I'm fun to be with, or I have good ideas"
- Draw conclusions based on findings
- Publish research
- Replicate study

- Quantitative Research methods falls into two "design" categories in psychology.
- Research methods that are **experimental** in design include the laboratory, field and quasi-experiment.

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Non-experimental methods include the observational, survey, interview and case study methods

- Experimental methods produce measurable quantitative data.

Non-experimental methods can sometimes give quantitative data but information is more likely to be descriptive or qualitative in nature.

Observation

- Perhaps the simplest form of research is (**Naturalistic) Observation**.
- It means, observing behavior in their natural environment. It often involves counting behaviors, such as number of aggressive acts, number of smiles, etc.

Correlational Studies

- Correlation means relationship, so the purpose of a correlational study is to determine if a relationship exists, what direction the relationship is, and how strong it is. **It can not make any assumptions of cause and effect** (no causation).

Experimental Studies

- Unlike correlational research methods or psychological tests, **experiments can provide information about cause-and-effect** relationships between variables.

In an experiment, a researcher manipulates or changes a particular variable under controlled conditions while observing resulting changes in another variable or variables

- **Variable:** A factor or element that can change in observable and measurable ways.
  - **Independent Variable (IV)** – the variable that is manipulated by the experimenter (**input variable**)-effects the experimenter wishes to examine.
  - **Dependent Variable (DV)** – the outcome variable (results of the experiment)-experimenter wants to find out if this variable depends on some other factor.
- **The control group:** made up of individuals who are randomly assigned to a group but do not receive the treatment. The measures taken from the control group are then compared to those in the experimental group to determine if the treatment had an effect.
- **The experimental group:** made up of individuals who are randomly assigned to the group and then receive the treatment. The scores of these participants are compared to those in the control group to determine if the treatment had an effect.
- **Ethics**

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In Psychology we are also interested in doing things ethically. This usually handled by the ethical review committee and ethical standards boards. The main aim here is to promote the followings by promoting individual dignity, human welfare and scientific integrity. It can be summarized as follows:

- Ensure no harm will come to subjects
- Informed consent
- Confidentiality
- Deception
- Debriefing

**These lecture notes are adaptations of several authors obtained from both textbooks and the internet sources.

Main Reference

Kemmer, Ibi-Agiobu. (Ed).(2009). *Essentials of psychology.* Lagos:

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